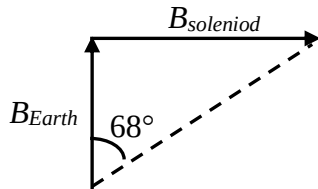


space. The induced e.m.f. is at 0.20 V, R will have 2.0 V across it while L will have 0.6 V across it.

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C

Assuming B_{solenoid} is to the right,



$$\tan \theta = \frac{B_{\text{solenoid}}}{B_{\text{Earth}}}$$

$$B_{\text{solenoid}} = B_{\text{Earth}} \tan \theta$$

$$= 2.0 \times 10^{-5} \tan 68^\circ = 4.95 \times 10^{-5}$$

$$B_{\text{solenoid}} = \mu n I$$

$$4.95 \times 10^{-5} = (4\pi \times 10^{-7}) \left(\frac{20}{15 \times 10^{-2}} \right) I$$

$$I = 0.30 \text{ A (2 s.f.)}$$

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B

Sliding the rod right or left will induce an e.m.f. across the rod. Since the rod is in the